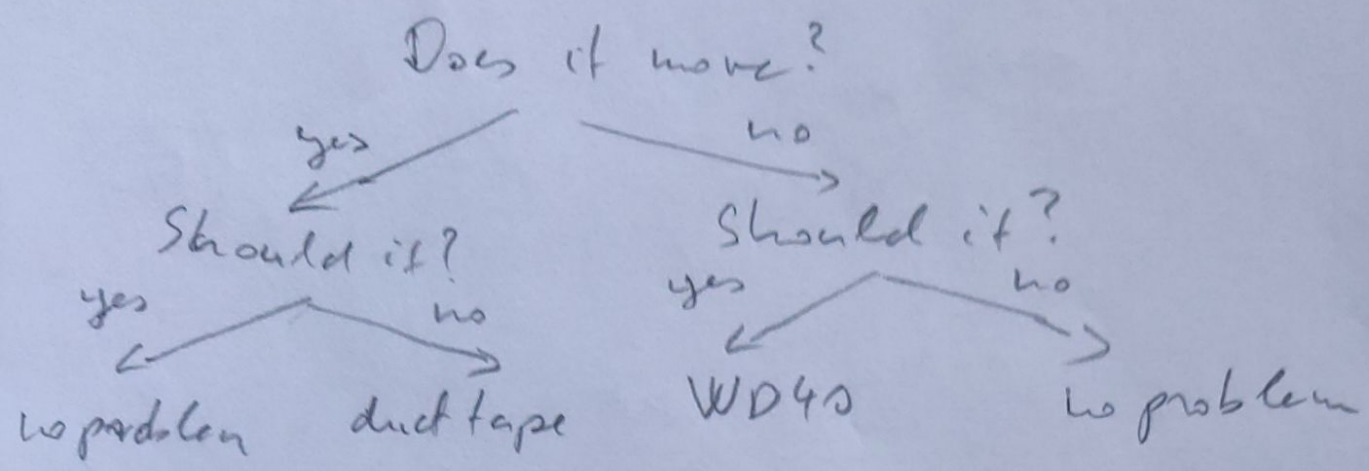
IML -2-

Decision tree



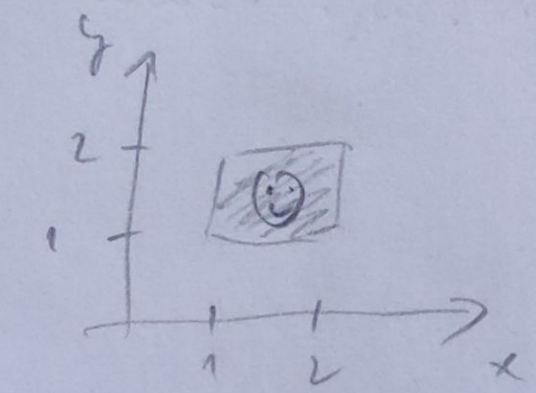
Outlook	Temp	Humidity	Wind	Tennis
Sunny	Hot 29,4	High	Weak	No
Sunny	Hot 26,6	High	Strong	No
Overcast	Hot 28,3	High	Weak	Yes
Rain	Mild 21,1	High	Weak	Yes
Rain	Cool 20,0	Normal	Weak	Yes
Rain	Cool 18,3	Normal	Strong	No
Overcast	Cool 17,7	Normal	Strong	Yes
Sunny	Mild 22,2	High	Weak	No
Sunny	Cool 20,6	Normal	Weak	Yes
Rain	Mild 23,9	Normal	Weak	Yes
Sunny	Mild 23,9	Normal	Strong	Yes
Overcast	Mild 22,2	High	Strong	Yes
Overcast	Hot 27,2	Normal	Weak	Yes
Rain	Mild 21,7	High	Strong	No

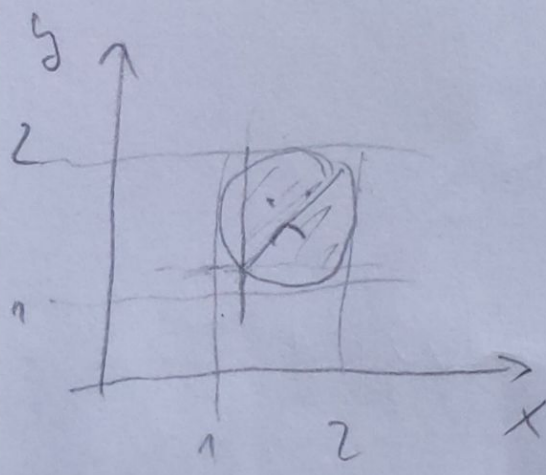
Task: create a decision tree, which tells us wether to play or not

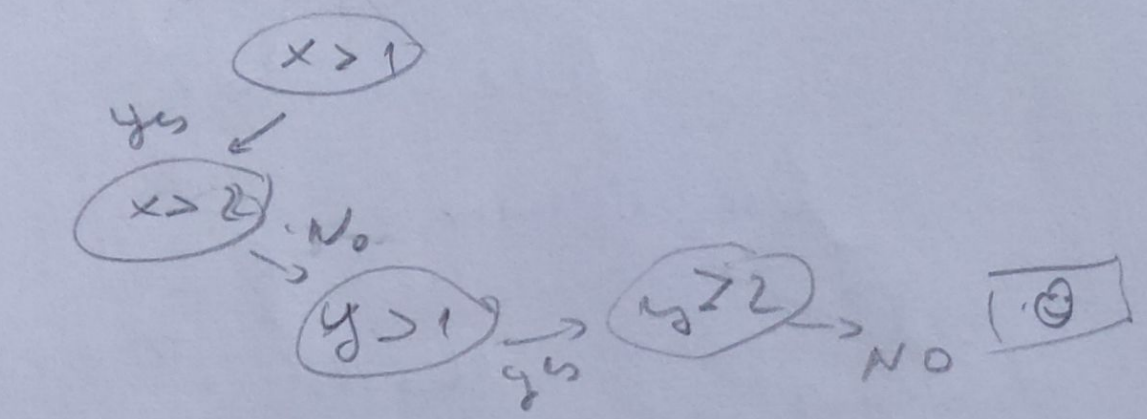
Questions: is Humidity High?

is Temp > 21?

Numerical value







→ $\left(x-\frac{3}{2}\right)^2 + \left(y-\frac{3}{2}\right)^2$

↳ irrelevant variables

→

When it is useful?
- small number of questions!

Ideally one single question

Which question to ask first?

We want to gain information!

What is information?

The measure of information is entropy

Entropy is the amount of uncertainty

$$H(X) = -\sum_{x \in X} p(x) \ln p(x)$$

If outcome is binary then $\log_2$ is used

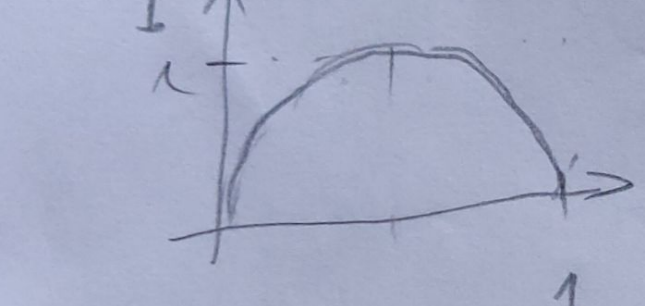
$$I(p(c_1), p(c_2) \dots p(c_k)) = -\sum_{i=1}^{k} p(c_i) \log_2(p(c_i))$$

- Full order: $p_1 = 1$ $p_i = 0$ $i > 0$

$\hookrightarrow 1 \cdot \log_2 1 = 0$ $\hookrightarrow 0 \cdot \log_2 0 = 0$

$I = 0$

- Full mix: $p_i = \frac{1}{k}$ $I = \log_2 k$ $\nexists\, k=2$ $I=1$



$p_2 = 1 - p_1$

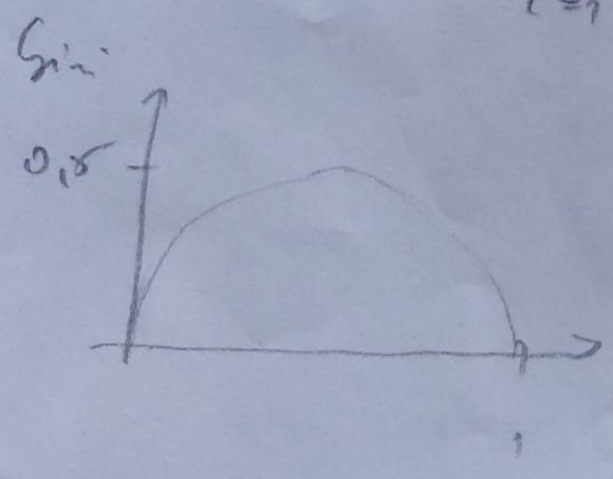
I: measure of uncertainty or surprise

$I = -\left(\frac{9}{14}\log_2\frac{9}{14} + \frac{5}{14}\log_2\frac{5}{14}\right) \approx 0,94$

Other measure: Gini - index

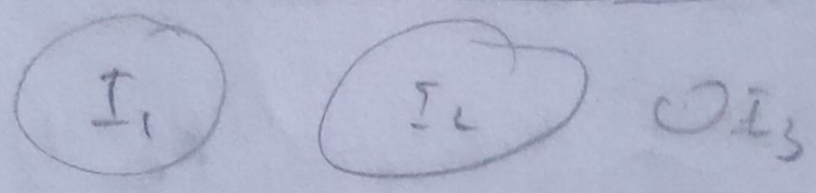
The probability of a random instance misclassified when chosen randomly

$$Gini = 1 - \sum_{i=1}^{k} p(i)^2$$



$Gini = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 \approx 0,46$

Information of multiple groups



$$\bar{I} = \sum_{\alpha} r_\alpha \cdot \bar{I}_\alpha$$

r_α : fraction of elements in group α

Information gain.

$$I_{gain} = \bar{I}_{before} - \bar{I}_{after}$$

Test Outlook:

Outlook = { Sunny : 2y, 3n
Overcast : 4y, 0n
Rain : 3y, 2n

$$I_{gain}(Outlook) = 0,94 - \left[\frac{5}{14}\bar{I}\left(\frac{2}{5},\frac{3}{5}\right) - \frac{4}{14}\bar{I}\left(\frac{4}{4},\frac{0}{4}\right) - \frac{5}{14}\bar{I}\left(\frac{3}{5},\frac{2}{5}\right)\right]$$
$$= 0,94 - 0,694 \approx 0,247$$

Humidity { Normal 6y 1n
High 3y 4n

$$I_{gain}(Humidity) = 0,94 - \left[\frac{7}{14}\bar{I}\left(\frac{6}{7},\frac{1}{7}\right) + \frac{7}{14}I\left(\frac{3}{7},\frac{4}{7}\right)\right] =$$
$$= 0,94 - 0,785 \approx 0,151$$

$I_{gain}(Temp) = 0,025$

$I_{gain}(Wind) = 0,048$

Winner : Overcast

Question for continuous data (python only bases this

- thresholds : only between values belonging to different classes

When to stop?

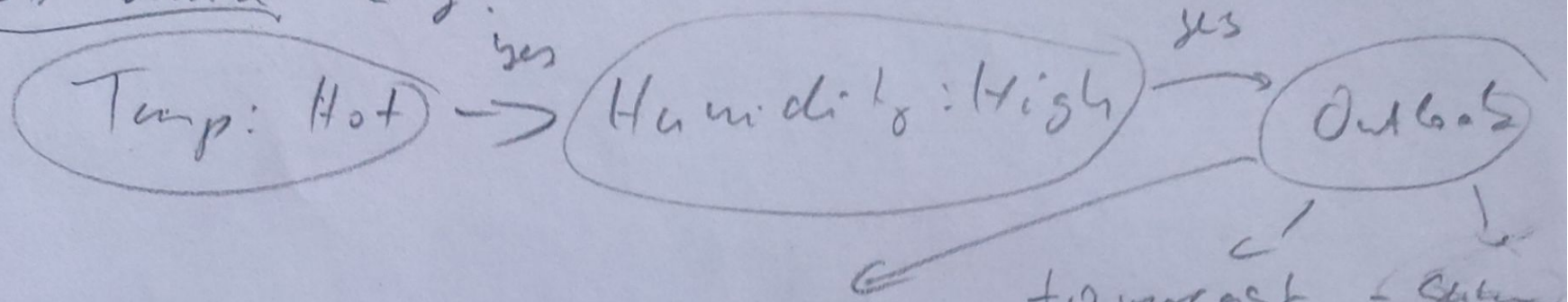
- All leaves have unique labels?
  - * Can be an enormous useless tree
  - * May not be possible with noisy data or missing feature

- Hyperparameter
  - * maximal depth
  - * minimal number of data in leaf
  - * minimum information gain
  - * predefined error

Missing Data

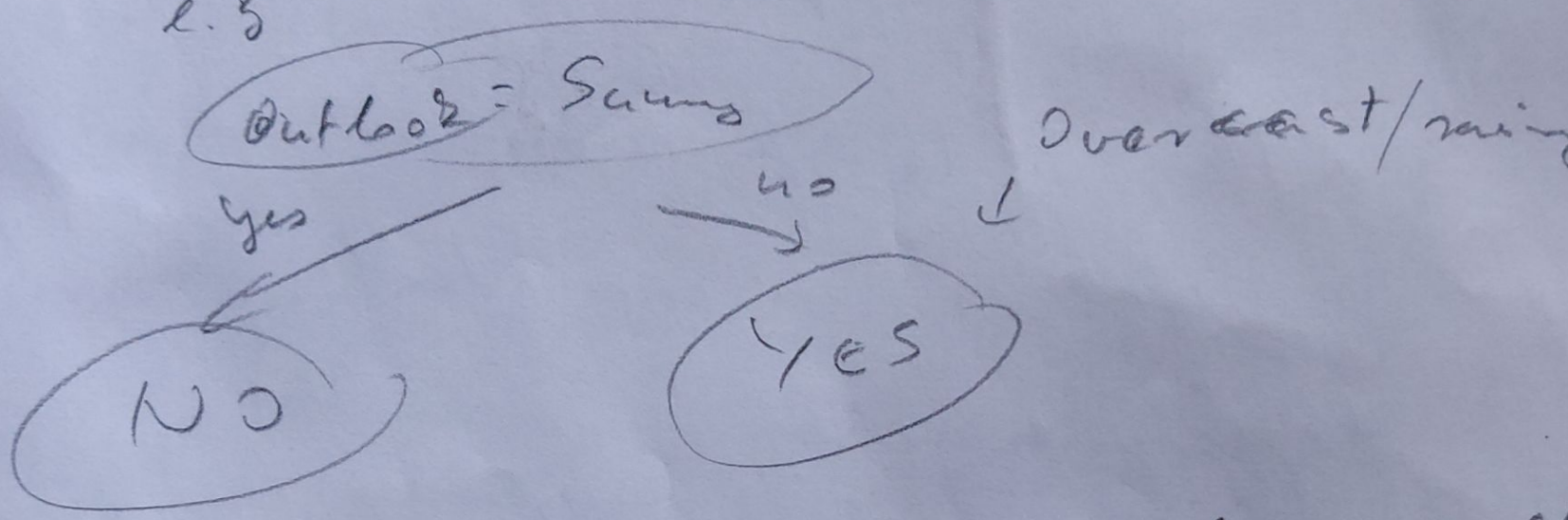
- average
- Separate category

No data e.g.



- Random decision
- In binary tree it never appears :

e.g



If numerically coded : result is random

Summary

What is a good tree?

- Not too big
  - ~ Computationally effective
  - ~ Human readable
- Not too small
  - ~ Meaningful decisions

pros/cons

+ Good for discrete values
+ Fast at test time
+ Straightforward interpretation
+ Parameter importance
+ Good with missing data

- Only for low dimensional data
- Boundary shape is extremely important